MICROSOFT · HANDS-ON LAB WORKSHOP

Agent Hosting on Azure

Three production patterns for running stateful AI agents on Azure

A Foundry Host Agent

B AKS + agent-sandbox

C ACA Sandbox

Deploy the same agent three ways · compare isolation, scale-to-zero, identity & cost · ≈135 minutes

How Today Works

Six short modules — one shared foundation, three solution labs, one wrap-up

0 **Introduction**

10 min · Framing & architecture — you are here

1 **Core Infrastructure**

30 min · Deploy once — shared by all three

2 **Solution A — Foundry Host Agent**

30 min · Managed on-ramp, deployed with azd

3 **Solution B — AKS + agent-sandbox**

40 min · Kata Container micro-VM isolation

4 **Solution C — ACA Sandbox**

20 min · Serverless, gVisor OS-level isolation

5 **Wrap-up & Decision Guide**

5 min · Compare, cost tips & Q&A

Before We Start

Prerequisites & setup check — fix any gaps now while we cover concepts

Accounts & Access

- Azure subscription — Contributor access
- Module 1 role assignments need Owner or User Access Administrator
- Role: Foundry User (for the hosted agent)
- Role: Container Apps SandboxGroup Data Owner

Tools to Install

- Azure CLI — az login completed
- Azure Developer CLI (azd) + Foundry extension
- kubectl and Docker (for Solution B)
- ACA preview extension (containerapp --allow-preview)

Most common blocker: creating role assignments needs Owner or User Access Administrator — plain Contributor can't. Confirm your access now.

Why Host Agents on Azure?

Production agents are stateful services — not stateless API calls

A demo agent is easy. A production agent must remember, isolate, authenticate, govern, and stay cheap when idle.

Isolation

Tenant / user boundaries

Scale-to-zero

No cost while idle

Identity & Auth

Managed identity, no secrets

Governance & Cost

One gateway, predictable spend

One size doesn't fit all — enterprise and consumer priorities differ. This workshop = three patterns for the same agent.

What Is a “Hosted Agent”?

The contract every solution in this workshop honors

Persistent context

Remembers the conversation across many turns

Managed-identity auth

Authenticates to the model with an Azure identity — no static secrets

Scale-to-zero + restore

Hibernates when idle, reloads state on the next request

All LLM calls via APIM

Every model call routes through one governed AI Gateway

Whatever the hosting technology, the agent behaves the same to a caller.

Two Target Scenarios

Enterprise vs. consumer — the personas that drive every design choice

ToB — Enterprise

Business / internal platforms

- Strong tenant & department isolation + audit
- Entra ID — SSO, RBAC, Conditional Access
- Data residency, private networking, compliance
- Reserved or burstable, predictable capacity

Security · Governance · Reliability

ToC — Consumer

Individual end-users / small teams

- Very large volume of short-lived sessions
- Process- or container-level isolation (lighter)
- Social login (External ID / B2C) or API key
- Aggressive scale-to-zero, pay-per-use

Cost · Speed · Simplicity

Three Solutions at a Glance

The whole day in one view — same agent, three hosts

Foundry Host Agent

BEST FOR

ToB managed · fastest on-ramp

ISOLATION

Managed (per-agent)

GA

AKS + agent-sandbox

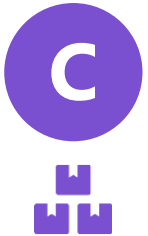
BEST FOR

ToB high-security

ISOLATION

Micro-VM (Kata)

GA



ACA Sandbox

BEST FOR

ToC / ToB long-running

ISOLATION

OS-level (gVisor)

Public Preview

What changes is isolation & control — Blob state and the APIM AI Gateway stay constant across all three.

Deploy Once: Core Infrastructure

Module 1 — the shared foundation all three solutions reuse

Shared platform

- Resource Group
- Blob Storage
- APIM (Basic v2)
- Key Vault
- Container Registry
- Entra App Reg
- User-Assigned MI

Foundry stack

- Foundry account
- Project maf-agent-prj
- gpt-5.4-mini
- Defender for AI
- 2 RAI policies
- APIM AI gateway

One deploy: `setup.sh → main.bicep → core.bicep` · Modules B & C discover it via the deploymentSN tag — reuse, never recreate.

The State Pattern

Azure Blob is the single source of truth — this is why scale-to-zero is free

New request

Load state from Azure Blob (per-agent JSON)

Active session

Save to Blob on every change (each turn)

Scale-to-zero

Nothing to flush — state already durable

Single source of truth: Azure Blob Storage. Each agent stores `<AGENT_ID>.json` in the `agent-state` container. No Redis / hot cache. Cool tier + versioning = cheap and recoverable.

Security: Identity + AI Gateway

No static keys anywhere — APIM is the one governed front door

Scope: user → agent authentication is out of scope today. We secure the agent → model hops.

Path 1 — all solutions

- Agent generates an Entra ID token
- → APIM AI API (validate-jwt)
- → APIM calls the LLM with its UAMI

Path 2 — Solution A only

- Hosted agent → its Foundry project
- (Foundry-assigned identity / Foundry User)
- → APIM AI gateway → LLM

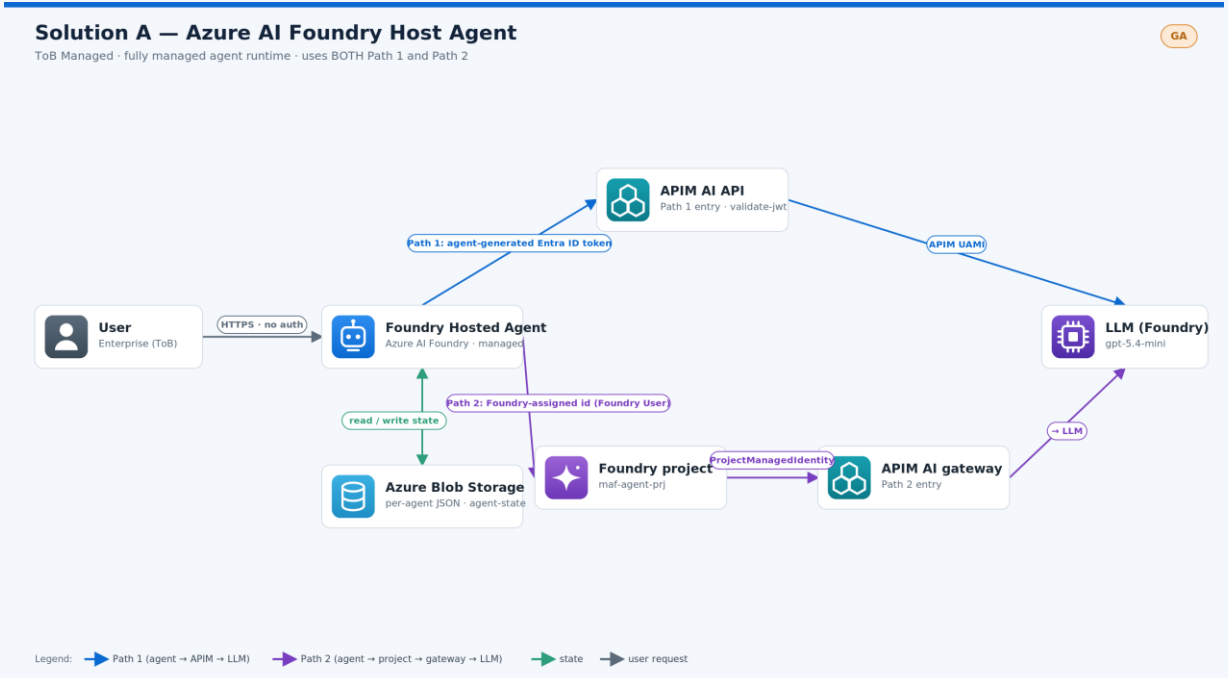
APIM = validate-jwt, rate limits, token quotas, logging, and it hides the Foundry endpoint.

Solution A — Foundry Host Agent

ToB Managed · fastest on-ramp · uses BOTH Path 1 and Path 2

AT A GLANCE

- Fully managed agent runtime (Azure AI Foundry)
- Deploy with azd — no new infra to stand up
- Model routing: direct or gateway (one env var)
- Runs on gpt-5.4-mini via the APIM AI Gateway
- Status: GA · Best for ToB managed



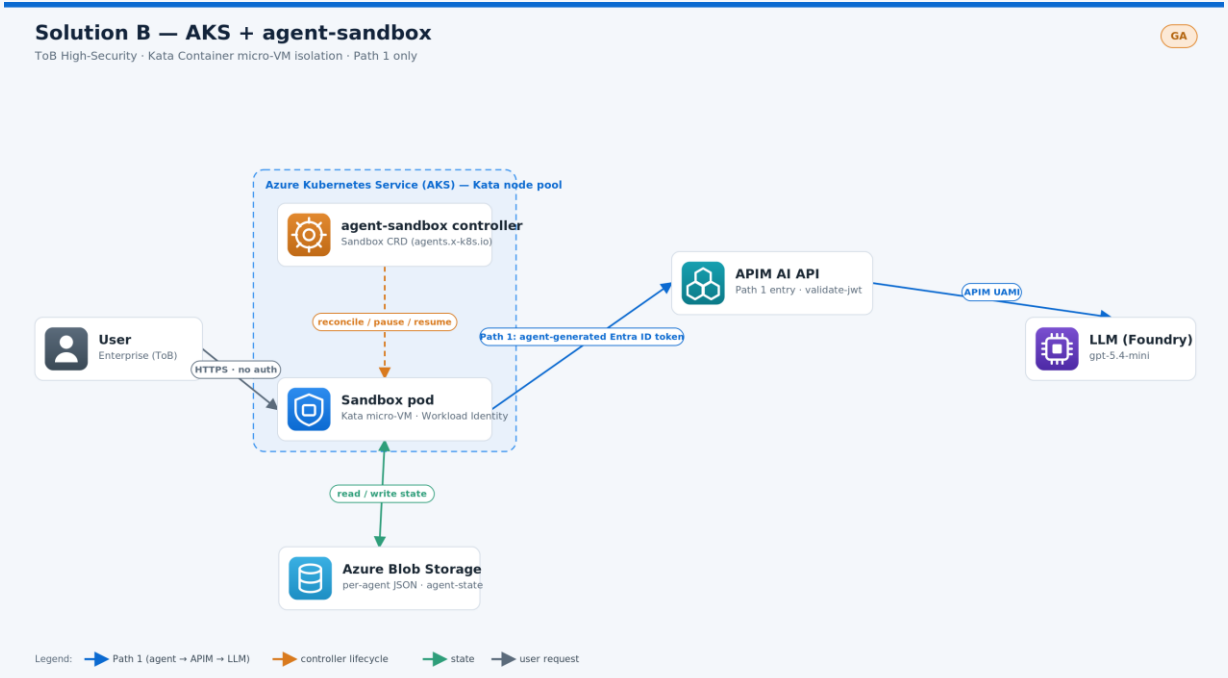
Architecture — see legend on the diagram

Solution B — AKS + agent-sandbox

ToB High-Security · Kata Container micro-VM isolation · Path 1 only

AT A GLANCE

- Strongest isolation — Kata Container micro-VMs
- AKS Pod Sandboxing (kata-vm-isolation)
- agent-sandbox controller: Sandbox CRD
- Lifecycle: pause / resume / hibernate
- Status: GA · Best for ToB high-security



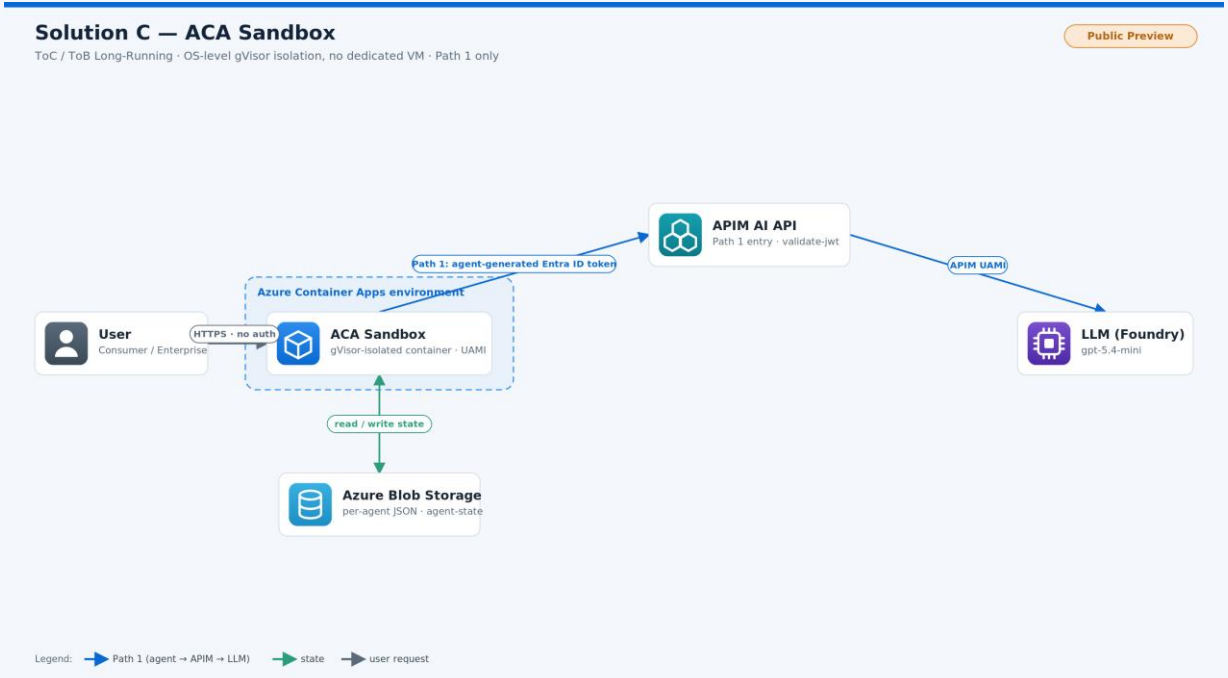
Architecture — see legend on the diagram

Solution C — ACA Sandbox

ToC / ToB Long-Running · gVisor OS-level isolation · Path 1 only (Public Preview)

AT A GLANCE

- OS-level isolation via gVisor — no dedicated VM
- Suspend / resume + snapshot state
- Reuses the Module 3 container image
- Per-agent container · UAMI / Workload Identity
- Status: Public Preview · ToC / long-running



Architecture — see legend on the diagram

Choosing the Right Solution

No universal winner — match the host to your operational profile

A	Fastest managed on-ramp for enterprise?	**Foundry Host Agent**
B	Maximum control, micro-VM isolation, private networking?	**AKS + agent-sandbox**
C	Long-running, strong isolation, serverless?	**ACA Sandbox**
→	One-time / untrusted code execution (code interpreter)?	**ACA Dynamic Sessions**

Full side-by-side comparison in Module 5 — isolation, scale-to-zero, state, Entra ID, APIM, ops complexity & cost.

Cost & Production Hardening

A look ahead to Module 5 — so the labs land in a production context

Cost levers

- Scale-to-zero after 30-min idle
- Blob Cool tier — ~50% cheaper
- Blob lifecycle rules expire stale state
- AKS Spot node pool — up to 90% off (B)
- Azure OpenAI PTU for high volume
- agent-sandbox WarmPool tuning (B)

Production hardening

- Private networking (VNet injection)
- Blob versioning + soft delete
- Azure Policy for compliance
- Monitor APIM 4xx / 5xx alerts
- Defender for Containers on AKS
- BCDR: verify state restores from Blob

Let’s build.

First stop → Module 1: deploy the shared foundation.

- Every module: one-command deploy or guided manual steps
- All scripts & READMEs are yours to keep
- Questions welcome throughout — total ≈ 135 minutes

```
$ cd agenthost/module-01  &&  ./setup.sh
```